



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

FINAL EXAMINATION JUNE 2023

Course: EC3114 (Software Modelling and Analysis)

Time: 8.00 am – 11.00 am
(3 hours)

Lecturer: Noor Zuhaili bt Md Yasin

Date: 6 October 2023

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of **3** printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Discuss **FOUR (4)** purposes of requirement engineering analysis. **(4 MARKS)**

b. State any **FIVE (5)** sections commonly found in software requirement specifications (SRS). **(5 MARKS)**

c. Explain **THREE (3)** phases of requirements engineering. **(6 MARKS)**

d. Discuss **FIVE (5)** interview fact-finding methods in requirements engineering. Include the criteria used during interviews and the types of information uncovered in interview sessions. **(10 MARKS)**
2. a. Explain the advantages of the following techniques for establishing user requirements: **(8 MARKS)**
 - i. Interviews.
 - ii. Questionnaires.
 - iii. Workshops.
 - iv. Document Sampling.
b. Explain the following terms used in Stakeholder Analysis: **(9 MARKS)**
 - i. Primary stakeholders.
 - ii. Secondary stakeholders.
 - iii. Tertiary stakeholders.
c. What does it mean to say that most modern development methods are iterative and incremental? **(5 MARKS)**

d. Discuss the difference between structural and behavioural diagrams as used in the Unified Modelling Language (UML). **(3 MARKS)**

(For Question 3 and Question 4 - Polo Travel Agency System)

3. Polo is a small travel agency based in England. They specialize in organizing and booking air trips to various remote and exotic destinations. They have access to a worldwide database of airlines and the corresponding flights. The main activities of the agency are:

i) Plan a new air trip for a specific customer. Each trip may include a few separate flights and possible excursions between flights.

ii) Book a trip. This involves reserving a seat on each flight and a place on any selected excursions. The customer must pay a deposit before the trip booking goes ahead. This booking must be confirmed within one week or cancelled.

In addition to the above, the agency needs to be able to amend the details of an air trip (including changing any reservations that have been made) or cancel the trip if necessary.

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| a. List the processes, data stores and external entities that you would include on a top-level Data Flow Diagram (DFD) of the Polo Travel Agency. You do not need to draw the DFD. | (10 MARKS) |
| b. Explain the different roles that Data Flow Diagrams (DFDs) and Entity Relationship Diagrams (ERDs) play in Structured Systems Development. | (10 MARKS) |
| c. Explain why DFDs are not normally used in Object Oriented Systems Development. | (5 MARKS) |

4. a. Consider the following extra information about the Polo system described above: **(10 MARKS)**

"The management of Polo wants to expand the system by introducing group trips in addition to individual trips. The following data should be stored about every individual trip: Trip No., Start date, End date, and Total cost. Group trips should also have two additional attributes: No of people, Discount."

Explain the following relationships between classes, using examples from the Polo system to illustrate your answers:

- i. Association.
- ii. Generalisation/Inheritance.

- b. Produce a sequence diagram for the use case 'Cancel a trip' in the Polo system described above. A brief description of this use case is given below. **(15 MARKS)**

"The trip's number is entered by an agent. The system searches the trip's details and relevant customer's details and displays both sets of details. Next, the details of all constituent flights are displayed. Finally, the system asks the user to confirm cancellation and the given trip is cancelled".

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